Chapter 1: Introduction

Course Notes

CSE0469 Blockchain: Special Topics in SW Engineering

Yusuf Altunel

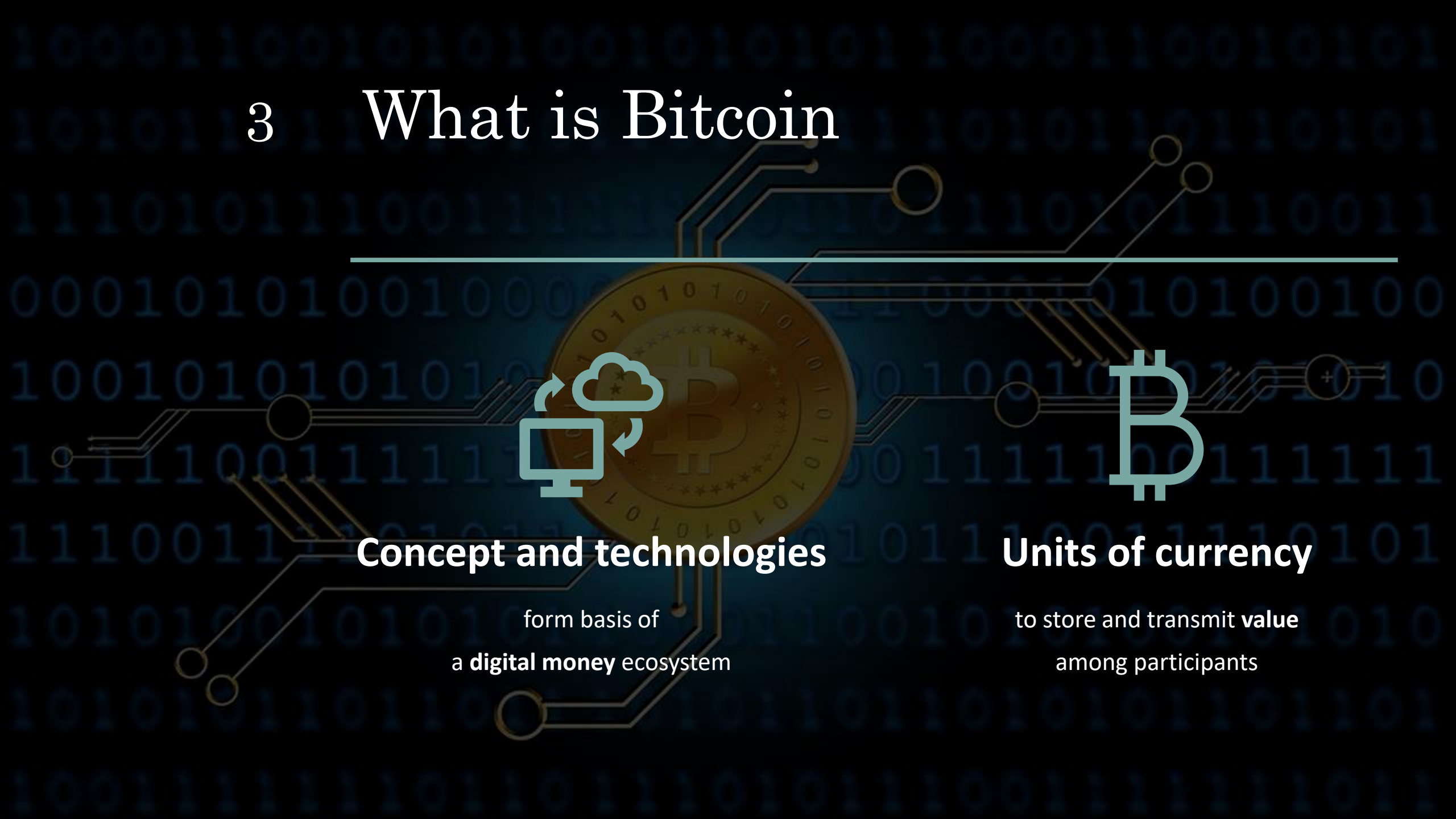
Based on Mastering Bitcoin 2e; Andreas M. Antonopoulos; O'Reilly, 2017

2 Content

- What is Bitcoin?
- History of Bitcoin
- Bitcoin Uses, Users, and Their Stories
- Getting Started



3 What is Bitcoin



Concept and technologies

form basis of

a **digital money** ecosystem

Units of currency

to store and transmit **value**

among participants

4 What is Bitcoin





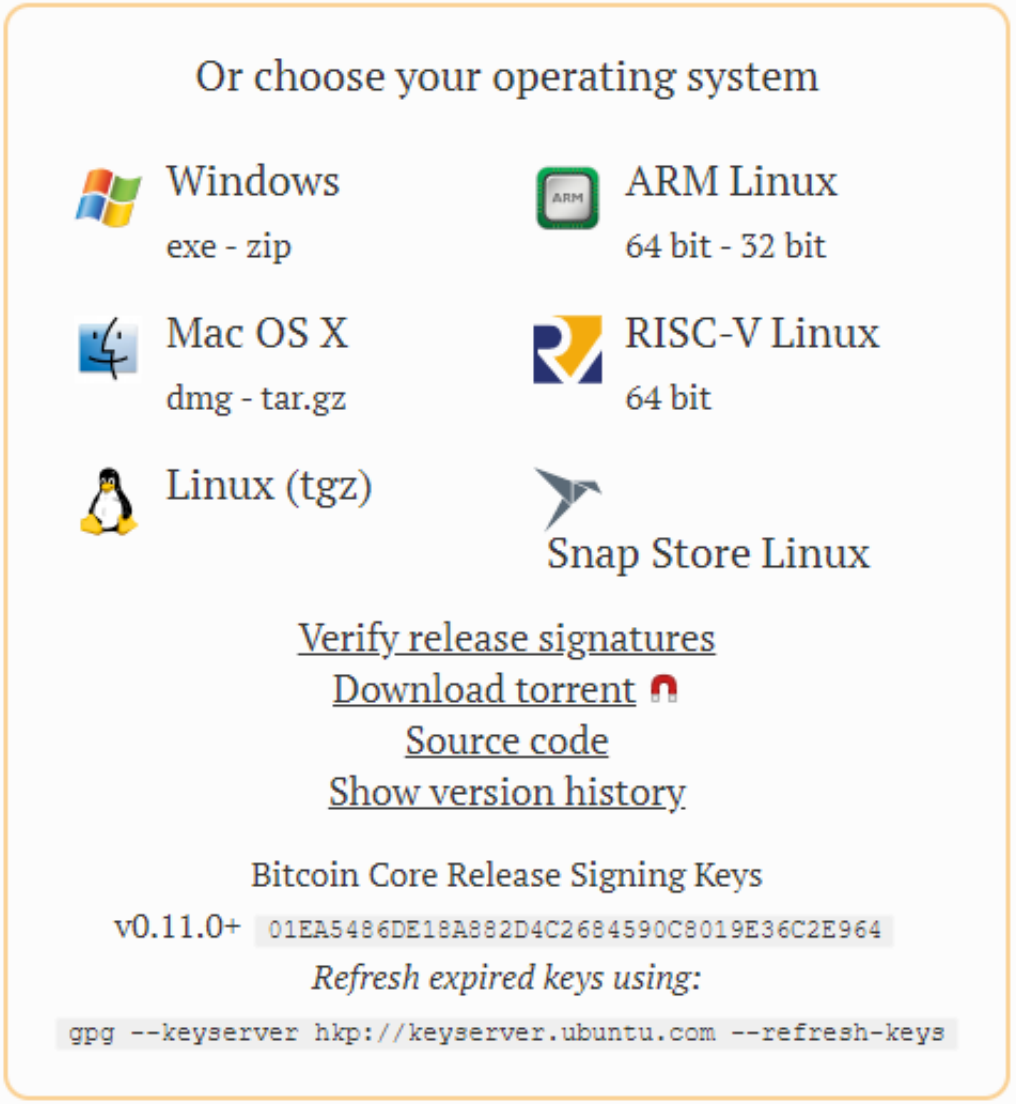
Protocol

Bitcoin users communicate with

bitcoin protocol

the internet or transport networks

What is Bitcoin



Bitcoin

open source software

available GitHub

Runs on various hardware

laptops

smartphones

Operating systems

MS Windows

LINUX

Mac OS X

6 What can be done with Bitcoin?



Transfer Bitcoin

Users can transfer bitcoin over the network

Conventional currency

buy and sell goods

send money to people or organizations

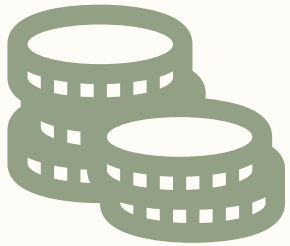
extend credit

Currency operations

Sell & Purchase

Exchange Bitcoin with other currencies

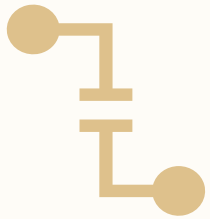
Basic characteristics



Virtual

No physical coins

Basic characteristics



Borderless

Fast, secure & borderless

Bitcoin in a sense is the perfect form of money for the internet because it is fast, secure, and borderless.

Basic characteristics



Distributed

peer-to-peer system

no “central” server or point of control

Basic characteristics



Decentralized

free of any central authority or point of control

avoid a single point of attack or corruption

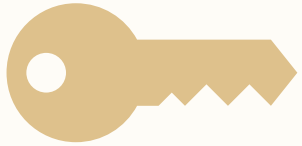
Basic characteristics



Robust

against intervention of legitimate governments or criminal elements

Ownership, Keys & Wallets



Ownership

Keys are the only instruments

to prove the ownership

Ownership, Keys & Wallets



Keys

Signing transactions



Signing

users can sign transactions to unlock the value



Spending

Transferring the ownership to a new owner.

Ownership, Keys & Wallets



Wallets

Digital wallet

Offline wallet



Digital wallet

on user's computer

or smartphone



Offline wallets

print

or physical forms

Mining

Mining

competing

to find solutions

to a mathematical problem

Coverage

processing bitcoin transactions

Validation

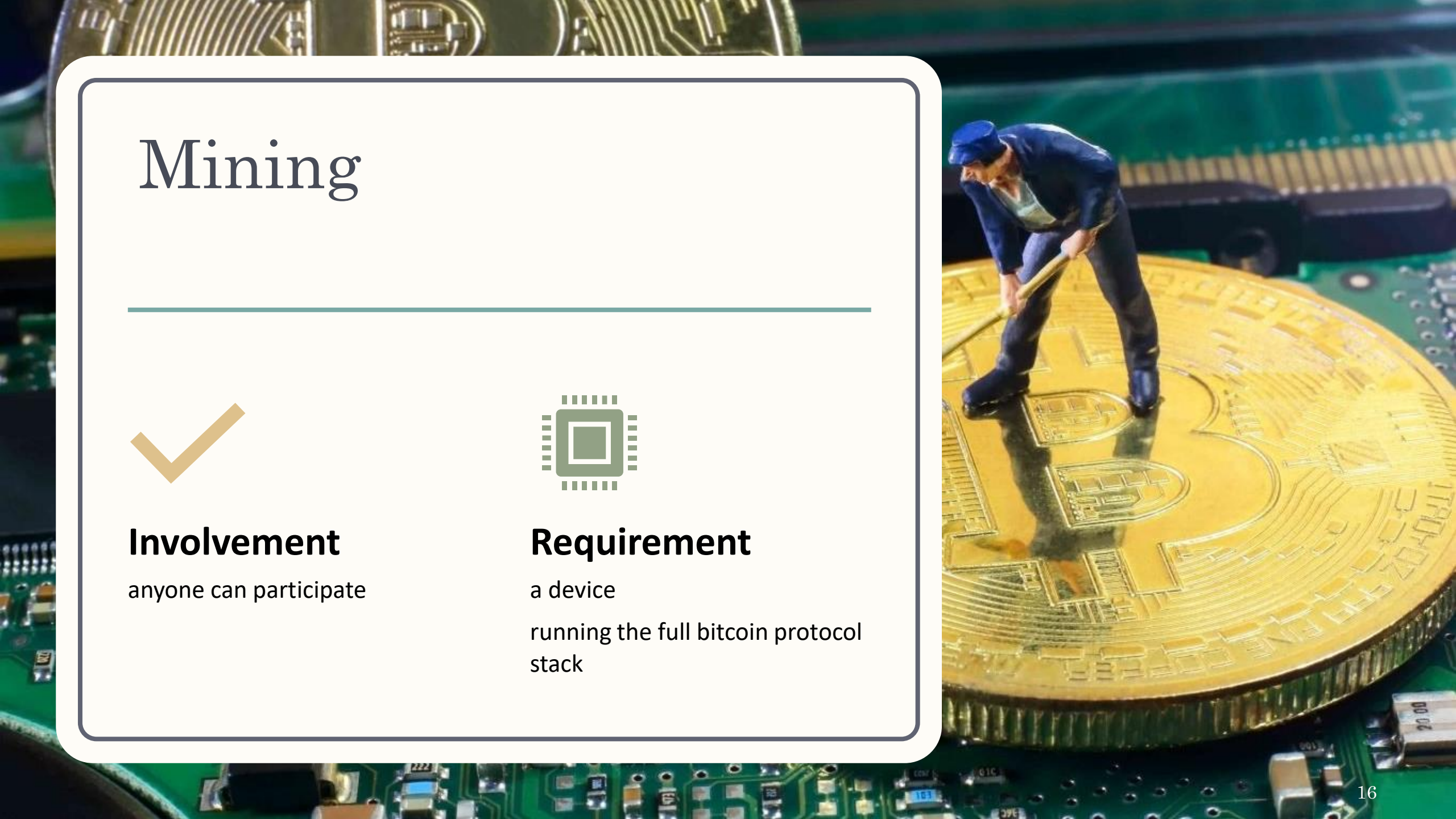
The transactions

are valid

are genuine



Mining



Involvement

anyone can participate

Requirement

a device

running the full bitcoin protocol stack

Mining

Duration

on average

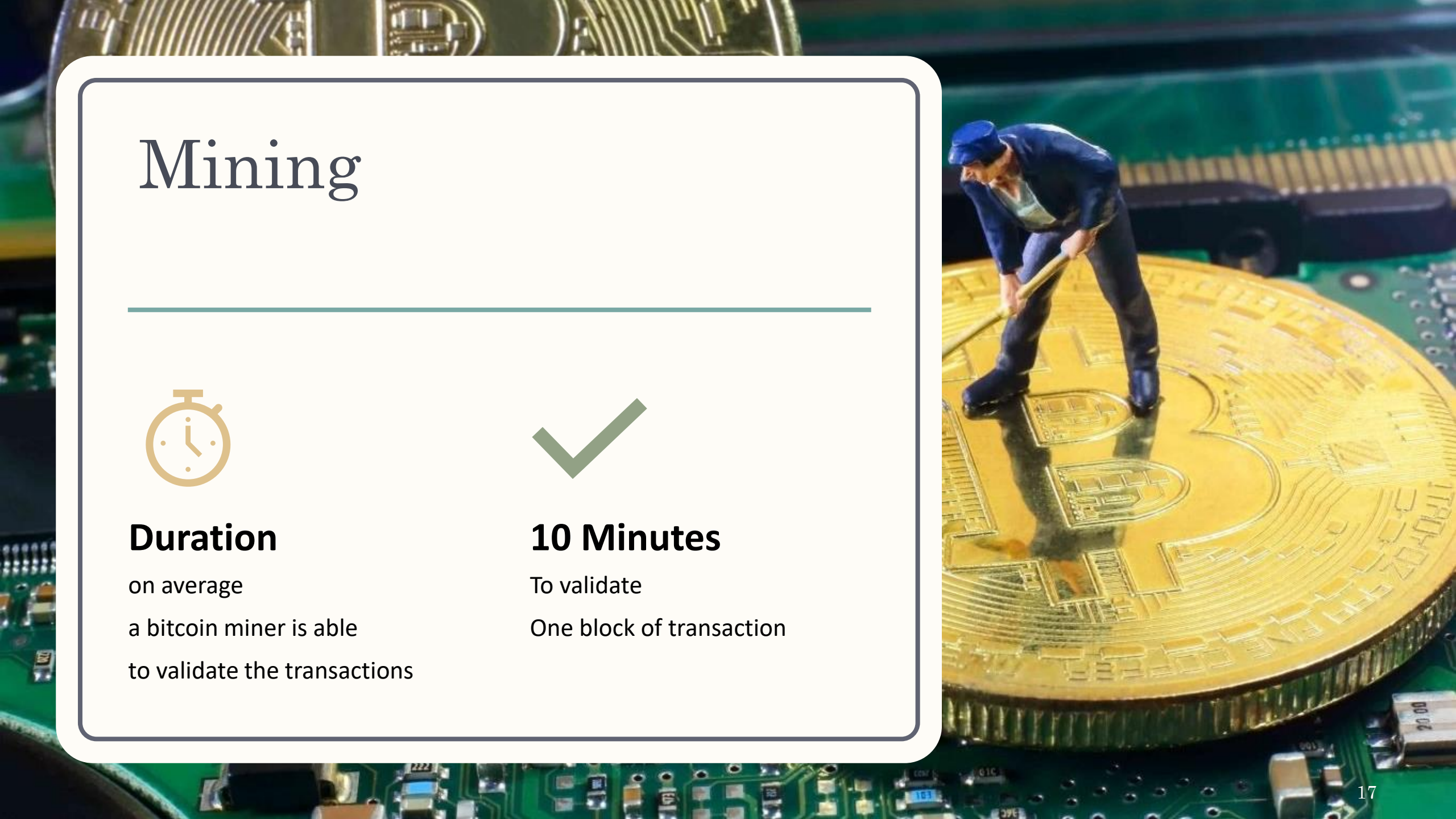
a bitcoin miner is able

to validate the transactions

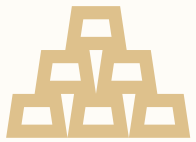
10 Minutes

To validate

One block of transaction



Circulation of Bitcoin



Minting

Bitcoin are created

through mining



Reward

Miner is rewarded

with brand new bitcoin

Circulation of Bitcoin



Decentralization

bitcoin mining

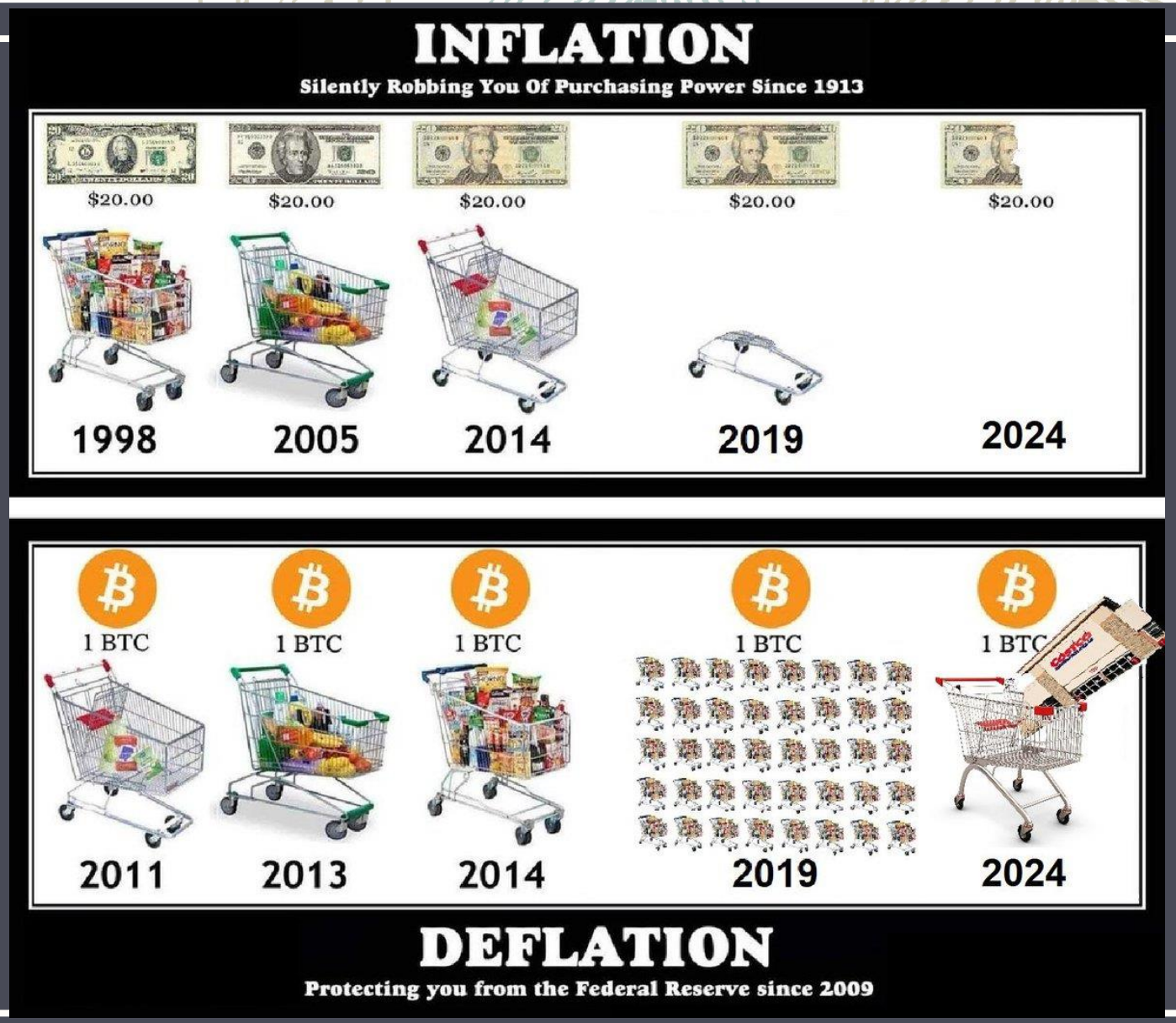
Decentralizes the currency-issuance

No central banks

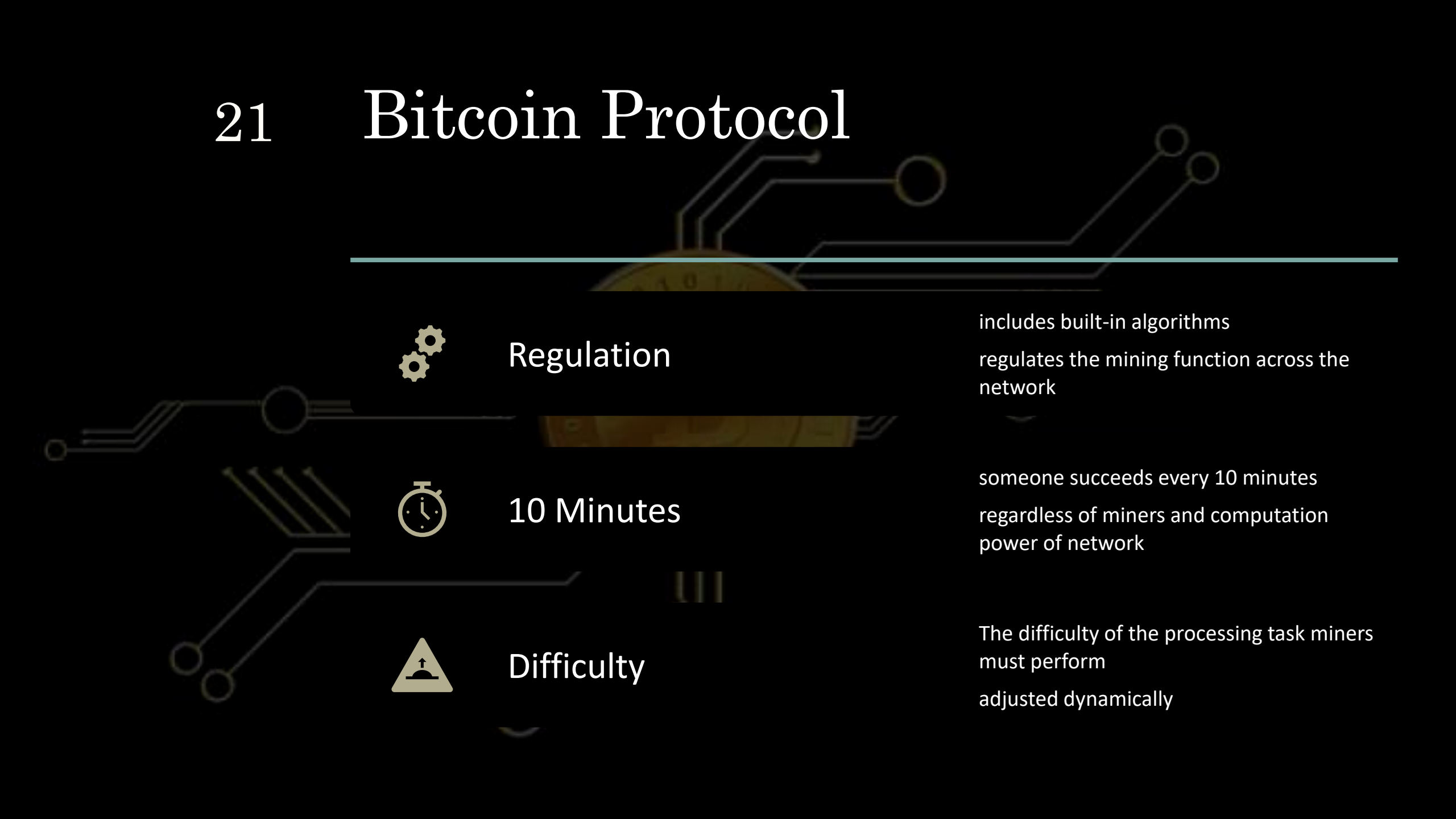
clearing functions and need

for any central bank.

Bitcoin is Deflationary



21 Bitcoin Protocol



Regulation

includes built-in algorithms

regulates the mining function across the network

10 Minutes

someone succeeds every 10 minutes

regardless of miners and computation power of network

Difficulty

The difficulty of the processing task miners must perform

adjusted dynamically

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Bitcoin Protocol

4 years period

The protocol halves the rate

every 4 years

new bitcoins are created

21M Coins

the total number of bitcoins

to be created fixed total

just below 21 million coins

2140

Bitcoin has diminishing rate of issuance

over the long term

to be completed

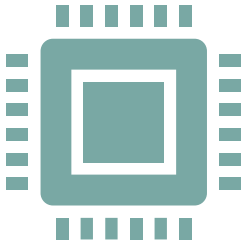
by the year 2140

Deflationary

Cannot be inflated by “printing” new money

above and beyond the expected issuance rate

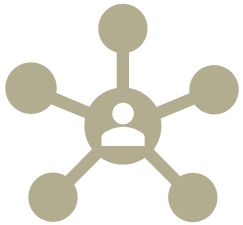
What is behind the scene?



4 key innovations

Brought together

A unique and powerful combination



4 key innovations

1. Bitcoin protocol
2. The blockchain
3. Consensus rules
4. Proof-of-Work algorithm



1. The bitcoin protocol

A decentralized peer-to-peer network



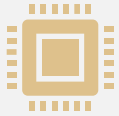
2. The blockchain

A public transaction ledger



3. Consensus rules

A set of rules for independent transaction validation and currency issuance



4. Proof-of-Work algorithm

A mechanism for reaching global decentralized consensus on the valid blockchain

What is behind the scene?

Bitcoin is

Internet of money

A network

- for propagating value and securing the ownership of digital assets
- via distributed computation

26 History of Bitcoin

HashCash, 1997, Adam Back

1997

1998

b-money, 1998, by computer scientist Wei Dai

Bitcoin is invented

- publication of a white paper
- "Bitcoin: A Peerto-Peer Electronic Cash System," written under the alias of Satoshi Nakamoto

2008

2009

The bitcoin network started in 2009

Proof-of-Work algorithm (mining) that provides security and resilience for bitcoin has increased in power exponentially

Satoshi Nakamoto withdrew from the public

developing the code and network to a group of volunteers

Apr. 2011

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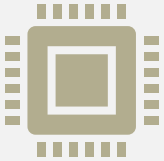
Nakotomo's Inventions



Early proposals

b-money, 1998, by computer scientist Wei Dai

HashCash, 1997, Adam Back



Bitcoin combines earlier concepts

Nakamoto combined several prior inventions

to create a completely decentralized electronic cash system

28 Nakotomo's Inventions

Specialities

- Decentralized
- Independent
- No authority
- Proof of Work
- Consensus
- 10 Minutes rule
- No double-spending

Who is Nakamoto?

Who is behind Bitcoin?

Still unknown

The identity

the person

people



Who is Nakamoto?

No control over Bitcoin

Neither Satoshi Nakamoto nor anyone else

exerts individual control over the bitcoin system

Operation based on

transparent mathematical principles

open source code

consensus among participants



Nakotomo's Inventions



No authority

does not rely on a central authority



Operations

currency issuance

settlement

validation of transactions

Nakotomo's Inventions



Proof-of-Work

The key innovation is

the "Proof-of-Work" algorithm



Benefit

Distribute

the validation of transactions

Nakotomo's Inventions



10 Minutes rule

conduct a global "election"



Period

every 10 minutes

Nakotomo's Inventions



Consensus

decentralized network



arrive at *consensus*

about

the state of transactions

Nakotomo's Inventions



double-spend problem

was a weakness of digital currency



was addressed

by clearing all transactions

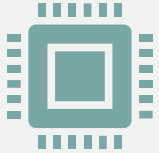
through a central clearinghouse

36 Getting Started

1. Choosing a Bitcoin Wallet
2. Quick Start
3. Getting Your First Bitcoin
4. Finding the Current Price of Bitcoin
5. Sending and Receiving Bitcoin

37 Choosing a Bitcoin Wallet: Platform

1. Desktop wallet
2. Mobile wallet
3. Web wallet
4. Hardware wallet
5. Paper wallet



Full-node client



Lightweight client



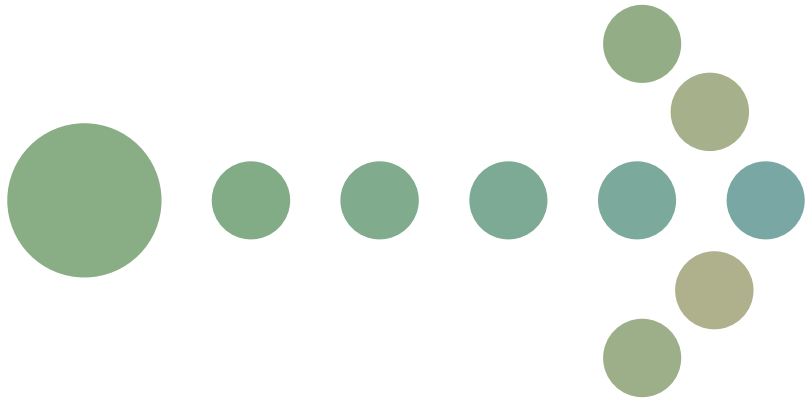
Third-party API client

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Bitcoin Wallet: Degree of Freedom

39 Full-node client

stores the entire history of bitcoin transactions



Lightweight client



SPV

known as a simple-payment-verification client



Connection to Full Nodes

connects to bitcoin full nodes

for access to bitcoin transactions



Storage

stores the user wallet locally and independently



Transactions

creates,

validates

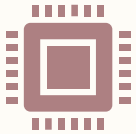
transmits transactions



Direct Interaction

interact directly with the bitcoin network, without an intermediary.

Third-party API client



Third party API

interacts with bitcoin

through a third-party system of

APIs: application programming interfaces



No direct connection

to the bitcoin network directly



Wallet storage

The wallet may be stored

by the user or

by third-party servers



Transactions

All transactions go

through a third party



Getting Your First Bitcoin



Finding the Current Price of Bitcoin



Sending and Receiving Bitcoin

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Quick Start

43 Wallet



Wallet

A wallet is simply

a collection of addresses and the keys

to unlock the funds within



Bitcoin address

Bitcoin addresses start with a 1 or 3

1Cdid9KFAaatwczBwBttQcwXYCpvK8h7FK.



Usage

Can be shared with other bitcoin users

to send bitcoin directly to your wallet.

Wallet

New Bitcoin Adresses	you can create new addresses as often as you like to direct funds to your wallet
QR code	Easy way to read the address to exchange via camera
Private keys	generated by the wallet Used together with bitcoin addresses

Getting Your First Bitcoin

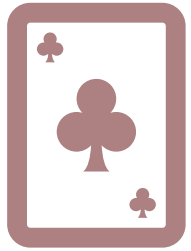


No traditional methods applicable

You cannot buy bitcoin

with your pay pal or credit card directly

Since account transfers are reversible



No credit card!

As a new user you cannot buy bitcoin instantly with a credit card

How to get bitcoin?



Buy from Friend

Find a friend who has bitcoin

and buy some from him or her directly



Find sellers

Use a classified service such as localbitcoins.com

to find a seller in your area to buy bitcoin for cash in an in-person transaction



Sell Product

Earn bitcoin by selling a product or service for bitcoin.

How to get bitcoin?



Use Bitcoin ATM

Use a bitcoin ATM in your city if any is available



Use Bitcoin Exchange

Use a bitcoin currency exchange linked to your bank account.



Local exchanges

Many countries now have currency exchanges

offer a market for buyers and sellers

to swap bitcoin with local currency

Finding the Current Price of Bitcoin



Exchange rate

Bitcoin, like most other currencies,

has a *floating exchange rate*.



Websites

There are hundreds of applications and websites

that can provide the current market rate.

Example:
https://www.coindesk.com/



Bitcoin Average

provides a simple view of

the volume-weighted-average for each currency.

https://bitcoinaverage.com/



CoinCap

A service listing the market capitalization and exchange rates

of hundreds of crypto-currencies, including bitcoin

https://coincap.io/



CME

Chicago Mercantile Exchange Bitcoin Reference Rate

A reference rate that can be used for institutional and contractual reference

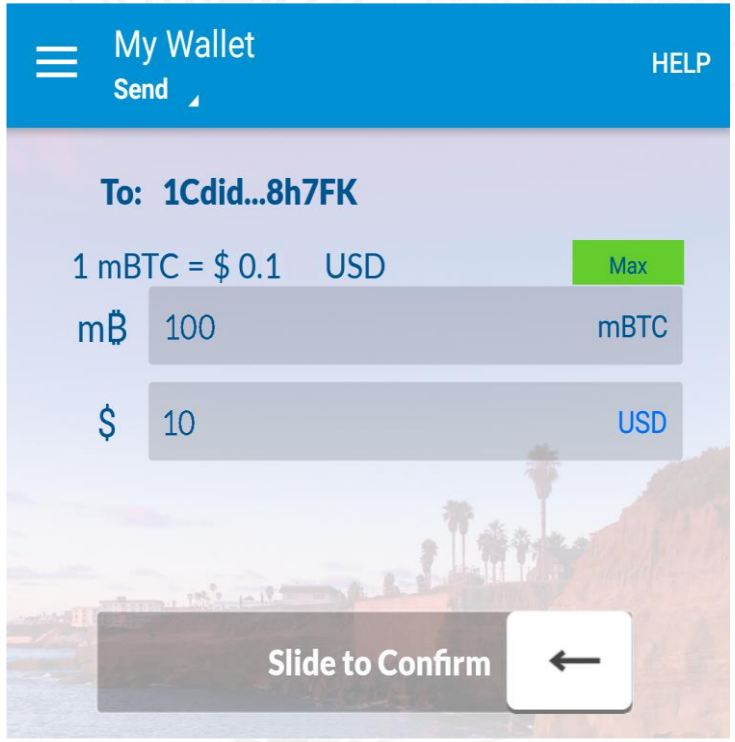
provided as part of investment data feeds by the CME

https://www.cmegroup.com/trading/cryptocurrency-indices.html

Sending and Receiving Bitcoin

Exchanging $10

- Alice has decided to exchange $10 US dollars for bitcoin,
- so as not to risk too much



Open Wallet

- She opens her Mycelium wallet application,

Receive

- When pressed receive
- Wallet displays a QR code with Alice's first bitcoin address.

Send from Joe

- Joe then selects Send on his smartphone wallet
- presented with a screen containing two inputs:

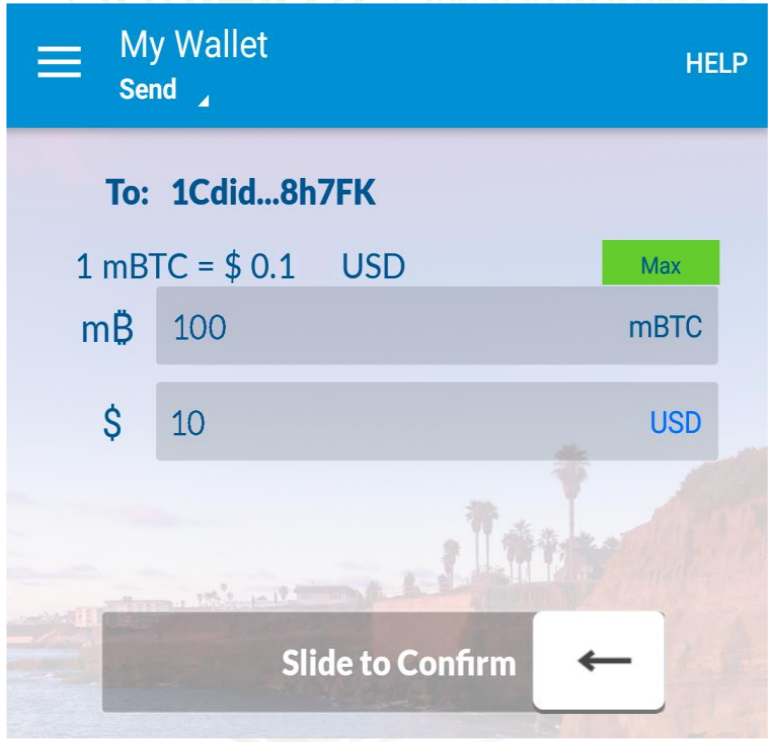
Enter Details

- A destination bitcoin address
- The amount to send, in bitcoin (BTC) or his local currency (USD)

50 Sending and Receiving Bitcoin

Exchanging $10

- Alice has decided to exchange $10 US dollars for bitcoin,
- so as not to risk too much



Read QR Code

- Joe taps the QR code icon and activates the smartphone camera,
- scanning the QR code displayed on Alice's smartphone

Enter Amount

- Joe enters the amount as $10 US

Calculate Exchange Rate

- The wallet converts it by accessing the most recent exchange rate from an online service.
- The exchange rate at the time is $100 US dollars per bitcoin,
- so $10 is worth 0.10 bitcoin (BTC), or 100 millibitcoin (mBTC)



https://www.youtube.com/watch?v=0gNPmLljJ1k



https://www.youtube.com/watch?v=7maNVTM8B1E



https://www.youtube.com/watch?v=fAmdC64KKdM

Discussions

Asses what you got so far!

Noticible issues	Interesting Details	Unclarified issues
What do you think on this chapter? Do you think the bitcoin inspiring?	What is the most interesting point? Select and explain a critical issu in this chapter	What are the points need further explanations?

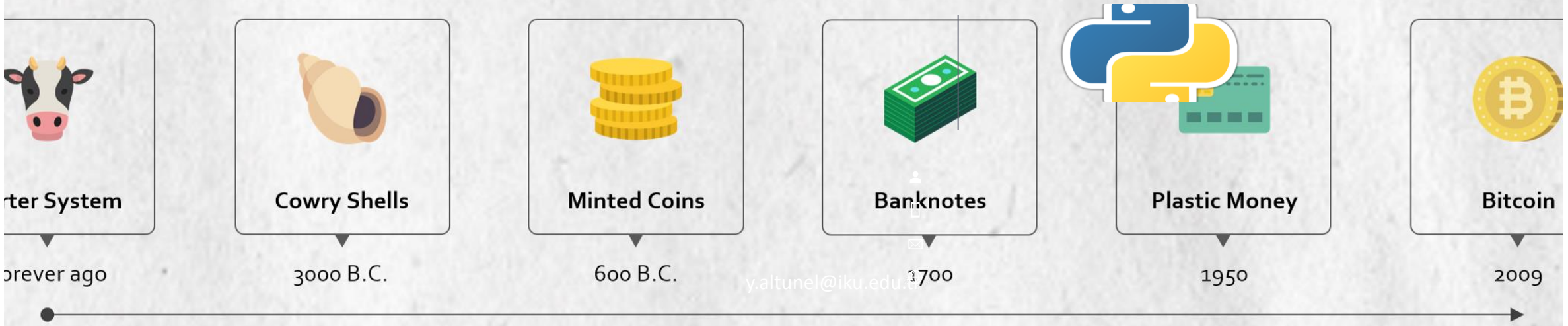






Thank You

The History of Money



Yusuf Altunel, PhD

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